Ziyang Liu

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EDUCATION

Soochow University | School of Future Science and Engineering *Sep 2022 – Jun 2026*
BSc in Statistics, GPA: 3.8/4.0 (90/100)
Core Courses: Fundamentals of Statistics, Probability Theory and Mathematical Statistics, Qualitative Data Analysis, Statistical Computing & SAS Software, Applied Multivariate Analysis, Applied Regression Analysis, Applied Time Series; Information Technology, Program Design and Application, Python Programming, Statistical Machine Learning.

Joint Program: Tsinghua University - X-Institute | Artificial Intelligence *Sep 2024 – Sep 2026*
Enhanced Student Research Training, Open Research for Innovation Challenges, Summer Research Training

PUBLICATIONS

[1] **Liu, Z.**, Chu, P., Dong, S., Zhang, C., Li, M., & Wang, J. (2025). FSRF: Factorization-Guided Semantic Recovery for Incomplete Multimodal Sentiment Analysis. In *Proceedings of the 2025 IEEE International Conference on Multimedia and Expo (ICME 2025, Oral)*

[2] Li, M., Hou, X., **Liu, Z.**, Yang, D., Qian, Z., Chen, J., Wei, J., Jiang, Y., Xu, Q., & Zhang, L. (2025). MCCD: Multi-Agent Collaboration-Based Compositional Diffusion for Complex Text-To-Image Generation. In *Proceedings of the 2025 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR 2025)*. https://doi.org/10.48550/arXiv.2505.02648

[3] Zhang, C., Wu, W., Zhang, Y., He, P., & **Liu, Z.** (2025). MERIT: Multimodal Emotion Recognition via RL-Enhanced Test-Time Adaptation. In *Proceedings of the 2025 International Conference on Machine Learning (ICML 2025): Affinity Workshop on New in Machine Learning (Virtual)*. https://icml.cc/virtual/2025/50449

RESEARCH

FSRF: Factorization-Guided Semantic Recovery for Incomplete Multimodal Sentiment Analysis [1]
Supervisors: Prof. Jin Wang, Prof. Min Cao | Soochow University *Dec 2023 – Mar 2025*

- Proposed a **Factorization-guided Semantic Recovery Framework (FSRF)** to tackle performance degradation in **Multimodal Sentiment Analysis (MSA)** under uncertain missing modalities.
- Developed a **De-redundant Homo-Heterogeneous Factorization (DHF)** module that decomposes multimodal inputs into **modality-homogeneous**, **modality-heterogeneous**, and **noise representations**, enabling more accurate disentanglement of sentiment-relevant information.
- Designed a **Distribution Alignment-based Self-Distillation (DAS)** mechanism utilizing **Sinkhorn distance** and **JS divergence** to achieve **bidirectional semantic recovery** and **distributional consistency** across missing modalities.
- Combined **representation factorization** and **hierarchical self-distillation** to enable fine-grained semantic recovery without reliance on complete modalities.
- Achieved **State-of-the-Art (SOTA)** performance on **CMU-MOSI** and **CMU-MOSEI** benchmarks, surpassing the previous **best model (CorrKD, CVPR 2024)** by **+2.09% (MOSI)** and **+1.88% (MOSEI)** in average F1-score under missing-modality conditions.

MCCD: Multi-Agent Collaboration-Based Compositional Diffusion for Complex Text-To-Image Generation [2]
Supervisor: Prof. Lihua Zhang | Fudan University *Jun 2024 – Mar 2025*

- Designed a **multi-agent collaboration framework (MCP)** that enables multiple agents to parse, reason, and refine complex text prompts for compositional diffusion generation.
- Integrated the framework with diffusion models (e.g., SDXL, SD2.0) to enhance spatial alignment and attribute binding **without retraining**, improving compositional control and realism.
- Achieved up to **9.0% performance gain** on **T2I-CompBench**, setting new SOTA results in complex scene generation.

Multi-Agent Reinforcement Learning Project

Supervisors: Prof. Bowen Zhou, Prof. Ning Ding | Tsinghua University *Apr 2025 – Jun 2025*

- Explored collaborative optimization and reinforcement learning of multi-agent systems under complex task settings.
- Constructed a **multi-agent interaction framework**, designed **task environments**, and implemented **coordination mechanisms** among agents.
- Combined **Test-Time Reinforcement Learning (TTRL)** with multi-agent training to enhance **policy generalization** and **adaptability**.
- Independently developed and debugged major code modules; assisted in theoretical derivation of TTRL convergence and structural analysis.

Self-Referential Evolution in Multi-Agent Systems

Supervisors: Prof. Liu Yang | National University of Singapore, Tsinghua University *Jun 2025 – Present*

- Investigating **self-referential iteration** and **continuous evolution** mechanisms enabling AI agents to achieve continuous improvement through **self-understanding** and **self-optimization**.
- Drew inspiration from biological evolutionary mechanisms to design two agent populations—**Task Agents** and **Optimizer Agents**—that co-evolve through mutual feedback.
- Developed a **self-evolutionary iteration framework** allowing agents to autonomously adapt to diverse task environments and improve performance over successive generations.

PROJECTS

Giiso-WeChat: Customizable Intelligent Customer Service Bot

Supervisor: Prof. Haitao Zheng | Tsinghua University *Oct 2024 – Feb 2025*

- Led the development of a WeChat-based intelligent customer service system, integrating Large Language Models (LLMs) with the GeWechat API to support automated text and voice interaction.
- Built a modular back-end framework using **Flask**, **RESTful APIs**, and **Docker**, solving message routing, session management, and large-scale multi-user concurrency issues.
- Implemented adaptive dialogue workflows and plug-in integration for external APIs, enabling personalized responses and **reducing human intervention time by over 60%** in customer support scenarios.

AI-Memory: Intelligent Memoir Writing System for the Elderly

Supervisors: Min Tang, Lichen Mao | Byte Dance *Oct 2024 – Apr 2025*

- Developed an AI-powered memoir generation system to help elderly users record and organize life stories through multimodal dialogue, combining **ASR (Automatic Speech Recognition)** and **LLMs** for real-time text generation.
- Designed the frontend interface and sharing module, enabling users to view, edit, and share their memoirs seamlessly, with a scalable architecture built on **Flask** and **PostgreSQL** for efficient deployment

HONORS

Mathematical Contest in Modeling (MCM): **Meritorious Winner**, Team Leader	2025
Baidu Star Programming Contest: **Gold Award**	2025
MaTiBei Programming Contest: **Gold Award** (5th in Jiangsu)	2025
National College Student Innovation Competition: **National Third Prize**, Team Leader	2024
Huashu Cup China Undergraduate Mathematical Contest in Modeling: **First Prize**, Team Leader	2024

SKILLS

Programming: Python (Advanced), MATLAB (Intermediate), R (Intermediate), C (Beginner)

Software: PyCharm, VS Code, Jupyter Notebook, Git, Docker, Postman, Anaconda, LangChain, Linux